

# High-Efficiency Octave Bandwidth Rectifier for Electromagnetic Energy Harvesting

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**Abstract**—This letter presents the design and implementation of a compact high-efficiency octave microwave rectifier. A key highlight is the novel segmented impedance matching method, a unique approach that expands the rectifier bandwidth. The diode reactance is initially regulated by a series short-ended microstrip line. Impedance-compensated structures, characterized by varying admittance properties across an extensive frequency range, partition the operating frequency band into two segments based on the input impedance, thereby minimizing impedance variation. Ultimately, the input impedance is matched by a novel triple-band matching network. An octave rectifier was fabricated and measured. Results demonstrate that the rectifier achieves over 50% efficiency over 1.3–2.55 GHz (fractional bandwidth 64.9%) at 0-dBm RF input power. Even with a decrease in input power to −10 dBm, the rectifier maintains over 30% efficiency.

**Index Terms**—Broadband, energy harvesting, high efficiency, impedance match, rectifier.

## I. INTRODUCTION

ENERGY harvesting, as one of the new technologies, has a very promising application prospects [1], [2], [3]. Electromagnetic energy harvesting enables the construction of self-powered or near-self-powered systems, which are critical for applications, such as the Internet-of-Things devices [4], [5], wireless sensor networks [6], [7], [8], and communication nodes in remote areas. Broadband high-efficiency rectifiers that can be applied at low power are the focus of research for better harvesting of electromagnetic energy in space.

Several proposals have been suggested to enhance the operational efficiency of the rectifiers. Nonuniform transmission lines are used for impedance matching of wideband rectifiers to achieve high rectification efficiency [9], [10]. On the other hand, the rectifiers using harmonic controlling circuits have been successfully enhanced, expanding the bandwidth without increasing the size [11], [12]. In terms of novel impedance compression techniques, the primary method to achieve small size and high efficiency is to compress the variation range of the rectifier input impedance [13], [14], [15], [16], [17]. Using

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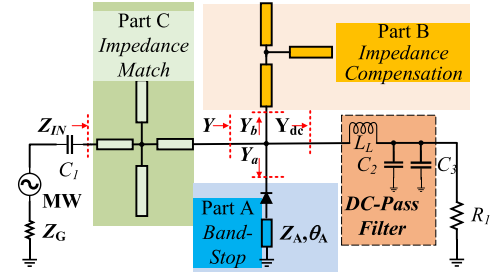


Fig. 1. Schematics of octave rectifier.

voltage-doubling circuits combined with broadband matching networks, such as simplified real-frequency techniques [18], [19], [20] and two-branch impedance matching circuits [21], can help make rectifiers compact and efficient.

This letter introduces an innovative octave rectifier with high efficiency for energy harvesting applications. The design includes impedance adjustment, compensation, and matching, which divides the rectifier's working frequency band into two segments: real impedance matching and complex impedance matching, improving both efficiency and operating bandwidth. We analyze the matching network and imaginary part compensation and then fabricate and measure an octave rectifier with an operating bandwidth of 1.3–2.55 GHz.

## II. PRINCIPLE AND DESIGN

Fig. 1 shows the operating principle diagram of the octave rectifier. It has three main parts: Part A adjusts the diode's admittance to  $Y_a$ , Part B supplies admittance  $Y_b$  to compensate  $Y_a$  to  $Y$ , and Part C matches the admittance  $Y$  to the source.  $Y_{dc} = 0$ , and  $Y = Y_a + Y_b$ .

In the design, to enhance the rectifier's bandwidth, the input admittance of the diode is optimized using two segments in its operating band. Define the starting and stopping frequencies of the operating band and the frequencies of the segments as fixed  $f_1$ ,  $f_3$ ,  $f_2$ , respectively, and  $f_1 < f_2 < f_3$ . It is required that the admittance of Parts A and B in parallel is complex from  $f_1$  to  $f_2$  and real from  $f_2$  to  $f_3$ , which meets the design requirement of segment matching. Part C comprises a cross structure intended for impedance matching. The operating bandwidth of the rectifier is effectively broadened by matching the input impedance of the rectifier in two segments. The electrical lengths are both defined at  $f_1$ .

### A. Impedance Compensation Structures

The topology of impedance compensation structures is shown in Fig. 2(a).  $Y_{b1}$  and  $Y_{b2}$  are the input admittances of microstrip lines. We denote  $k_2 = f_2/f_1$  and  $k_3 = f_3/f_1$ .

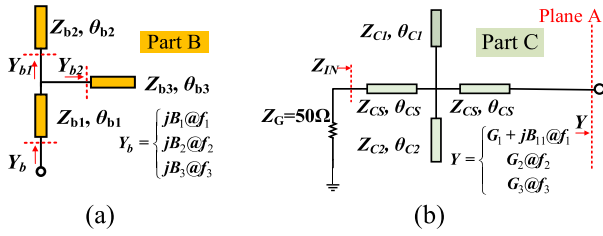


Fig. 2. (a) Impedance compensation structures. (b) Triple-band matching network.

To simplify the procedure, first, it is satisfied with  $Y_{b2} = 0@f_2$ , which could be achieved by implementing a quarter-wave open-ended transmission line. Subsequently, upon selecting and fixing the characteristic impedance of  $Z_{B1}$ , the electrical length of  $\theta_{b1}$  is expressed as follows:

$$\theta_{b1} = \frac{1}{k_2} \tan^{-1} \left( \frac{-1}{Z_{b1} B_2} \right). \quad (1)$$

When  $Z_{b1}$  and  $\theta_{b1}$  are calibrated, the admittance at  $f_1$  and  $f_3$  is

$$\begin{aligned} Y_{b1} + Y_{b2} &= \left[ \frac{-j(B_1 Z_{b1} + \tan \theta_{b1})}{Z_{b1} + Z_{b1}^2 \tan \theta_{b1}} \right]_{f=f_1} \\ &= \left[ \frac{-j(B_3 Z_{b1} + \tan(k_3 \theta_{b1}))}{Z_{b1} + Z_{b1}^2 B_3 \tan(k_3 \theta_{b1})} \right]_{f=f_3}. \end{aligned} \quad (2)$$

It also satisfies

$$\begin{aligned} Y_{b1} + Y_{b2} &= \left[ \frac{j \tan(\theta_{b2})}{Z_{b2}} + \frac{j \tan(\theta_{b3})}{Z_{b3}} \right]_{f=f_1} \\ &= \left[ \frac{j \tan(k_3 \theta_{b2})}{Z_{b2}} + \frac{j \tan(k_3 \theta_{b3})}{Z_{b3}} \right]_{f=f_3}. \end{aligned} \quad (3)$$

Solving (3) and (4) in association, the structural parameters of the entire Part B can be calculated.

### B. Impedance Match Structure

A novel triple-band matching network is proposed. As depicted in Fig. 2(b), the triple-frequency matching network comprises a cross-shaped microstrip network structure. When working with three frequencies,  $f_1$ ,  $f_2$ , and  $f_3$ , it can realize impedance matching at  $f_2$  and  $f_3$  for pure resistance as well as realize impedance matching at  $f_1$  for complex impedance.

A quarter wavelength transmission line is used as the impedance converter in impedance matching. The relationship between input impedance  $Z_{in}$  and load impedance  $Z_L$  meets

$$Z_{in} = \frac{Z_0^2}{Z_L} \quad (4)$$

where  $Z_0$  is the transmission line characteristic impedance.

For frequencies  $f_2$  and  $f_3$ , the cross circuit is equivalent to a quarter-wave transmission line, so both have the same  $ABCD$  matrix.  $ABCD$  matrix for Part C is as follows (5) and (6), as shown at the bottom of the next page, where  $Z_{CS}$ ,  $Z_{C1}$ ,  $Z_{C2}$ ,  $\theta_{CS}$ ,  $\theta_{C1}$ , and  $\theta_{C2}$  are the characteristic impedance and electrical length of microstrip lines, as shown in Fig. 2(b).

The  $ABCD$  matrix of a  $\lambda_0/4$  transmission line is

$$\begin{bmatrix} A_0 & B_0 \\ C_0 & D_0 \end{bmatrix} = \begin{bmatrix} 0 & jZ_0 \\ j\frac{1}{Z_0} & 0 \end{bmatrix}. \quad (7)$$

| TABLE I<br>SPICE PARAMETERS OF AN SMS7630 DIODE |          |           |             |          |
|-------------------------------------------------|----------|-----------|-------------|----------|
| $B_V$                                           | $C_{j0}$ | $I_s$     | $R_s$       | $V_{bi}$ |
| 2.0 V                                           | 0.14 pF  | 5 $\mu$ A | 20 $\Omega$ | 0.34 V   |

When  $A = D = 0$ , we can obtain

$$\frac{1}{Z_{CS}} \left( \frac{\cos^2 \theta_{CS} - \sin^2 \theta_{CS}}{\sin \theta_{CS} \cos \theta_{CS}} \right) = \frac{\tan \theta_{C1}}{Z_{C1}} + \frac{\tan \theta_{C2}}{Z_{C2}}. \quad (8)$$

We substitute  $B$  and  $C$  into the chain matrix and obtain

$$Z_0 = Z_S \tan \theta_S. \quad (9)$$

Cross circuits have characteristic impedances  $Z_{02}$  and  $Z_{03}$  at  $f_2$  and  $f_3$  [22]

$$Z_{02} = Z_{CS} \tan k_2 \theta_{CS} \quad (10)$$

$$Z_{03} = Z_{CS} \tan k_3 \theta_{CS}. \quad (11)$$

Because the admittances at  $f_2$  and  $f_3$  looking to the right in Plane A are  $G_2$  and  $G_3$ , respectively, we have

$$Z_{02}^2 = \frac{Z_{IN}}{G_2}, \quad Z_{03}^2 = \frac{Z_{IN}}{G_3}. \quad (12)$$

Solving (10)–(12), we can solve for  $Z_{CS}$  and  $\theta_{CS}$ . At frequency  $f_1$ , the  $ABCD$  matrix looking to the right from Plane A is

$$\begin{bmatrix} A_{f1} & B_{f1} \\ C_{f1} & D_{f1} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} 1 & 0 \\ G_{IN} & 1 \end{bmatrix} = \begin{bmatrix} A + BG_{IN} & B \\ C + DG_{IN} & D \end{bmatrix} \quad (13)$$

where  $G_{IN} = 1/Z_{IN}$ . Therefore, at  $f_1$ , it obtains

$$Y_1 = \frac{C_{f1}}{A_{f1}} = \frac{C + DG_{IN}}{A + BG_{IN}} = G_1 + jB_{11}. \quad (14)$$

Based on the derivation, the designed triple-band matching network can realize pure resistance impedance matching at  $f_2$  and  $f_3$  and complex impedance matching at  $f_1$ .

### C. Design of the Octave Rectifier

The design uses an SMS7630 diode with SPICE parameters shown in Table I. First, Part A is a bandstop structure [23], which can reflect the high harmonics and adjust the impedance of the diode to simplify subsequent design. The impedance  $Z_a$  ( $Z_a = 1/Y_a$ ) curve of the diode after series connection with Part A at different frequencies is shown in Fig. 3(a). The frequency intersecting with the real axis of the Smith chart is as  $f_2$ ,  $f_2 = 2.05$  GHz,  $f_1 = 1.3$  GHz, and  $f_3 = 2.6$  GHz. Then, determining the admittances  $Y_a$  corresponding to the  $f_1$ ,  $f_2$ , and  $f_3$ , design and optimize the impedance compensation structure and matching network according to  $Y_a$ . The impedance variation curve  $Z$  ( $Z = 1/Y$ ) of the rectifier in parallel with Part B is shown in Fig. 3(b). It is seen that the input impedance of the rectifier is zero in the imaginary part from  $f_2$  to  $f_3$ , whereas it is negative in the frequency range of  $f_1$  to  $f_2$ . The real parts of the rectifier input impedance at  $f_2$  and  $f_3$  are around 120 and 280  $\Omega$ , respectively, which are brought into (10)–(12) to obtain  $Z_{CS} \approx 100$   $\Omega$  and  $\theta_{CS} \approx \lambda_g/11$ . After optimizing the design, the input impedance  $Z_{IN}$  of the rectifier after cascading with matching network Part C is shown in Fig. 3(c), and the rectifier is well matched with the signal source.

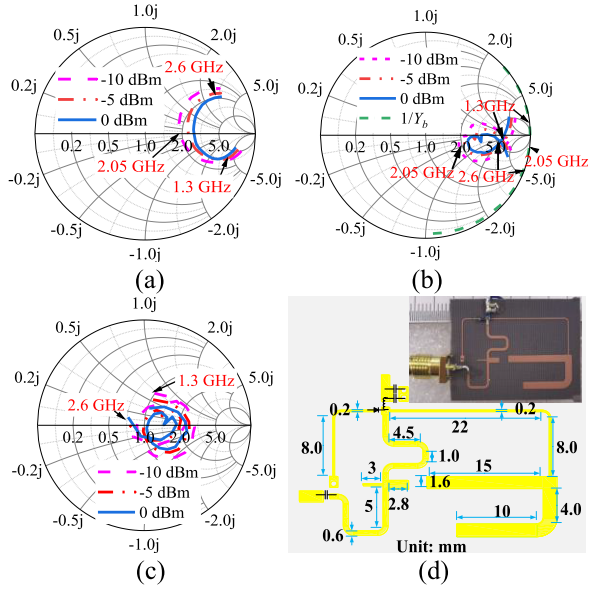


Fig. 3. (a)  $Z_a$ . (b)  $Z$  ( $Z = 1/Y$ ). (c)  $Z_{IN}$ . (d) Layout and photograph of the proposed octave rectifier.

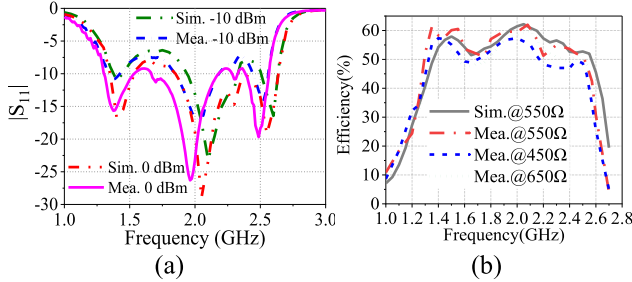


Fig. 4. (a)  $|S_{11}|$  with various frequencies. (b) PCE versus frequency with different dc loads at 0 dBm.

### III. MEASUREMENT AND ANALYSIS

We designed and fabricated an octave rectifier with an operating bandwidth of 1.3–2.6 GHz and measured its rectification efficiency. F4B-2 is the substrate for the rectifier, with a relative dielectric constant of 2.65. The layout and photograph of the octave rectifier are illustrated in Fig. 3(d).  $C_1$  (39 pF) is a dc-block capacitor. The dc-pass filter comprises an inductor  $L$  (22 nH) along with two capacitors,  $C_2$  (13 pF) and  $C_3$  (39 pF).

Fig. 4(a) displays the simulated and measured reflection coefficient  $|S_{11}|$  with different frequencies at 0 and 10 dBm. The  $|S_{11}|$  is less than -10 dB at 0 dBm, which is well matched. As depicted in Fig. 4(b), the conversion efficiency is simulated and measured against the frequency for an input power of 0 dBm with different dc loads. The rectifier achieves a PCE of more than 50% over 1.3–2.55 GHz with 550  $\Omega$ .

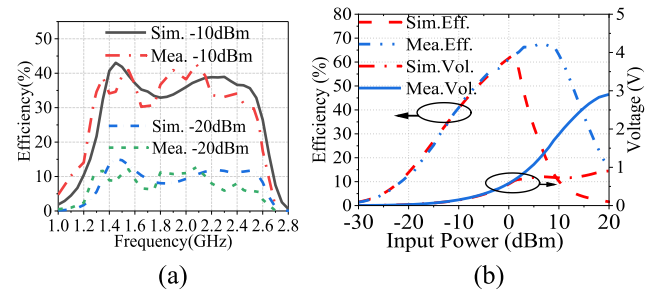


Fig. 5. (a) PCE at -10 and -20 dBm. (b) PCE versus input power at 2.08 GHz.

TABLE II  
COMPARISON WITH PREVIOUS RECTIFIERS

| Ref       | Diode Models and Number | Bandwidth range for PCE > 50% |                        |             | Size (mm <sup>2</sup> ) |
|-----------|-------------------------|-------------------------------|------------------------|-------------|-------------------------|
|           |                         | Frequency range (GHz)         | Relative bandwidth (%) | Power (dBm) |                         |
| [12]      | BAT15-03W, 1            | 1.77-2.88                     | 47.7                   | 0           | 23×23                   |
| [14]      | HSMS286, 1              | 2-3.3                         | 49.1                   | 4           | 31×18                   |
| [15]      | SMS7630, 2              | 1-1.7                         | 51.9                   | 0           | 35×25                   |
| [16]      | HSMS286, 1              | 1.5-2.85                      | 49.0                   | 5           | 37×10                   |
| [19]      | SMS7630, 4              | 1.6-3.4                       | 66.6                   | 8           | 22×24                   |
| This Work | SMS7630, 1              | 1.3-2.55                      | 64.9                   | 0           | 32×21                   |

It is reaching an octave level with a relative bandwidth of 64.9%. The measurement results at lower input power levels are shown in Fig. 5(a). At -10 dBm, the rectifier has a PCE of more than 30% in the octave range of 1.25–2.5 GHz. Fig. 5(b) depicts the rectifier operating optimally at 2.08 GHz, obtaining a maximum rectification efficiency of 67.5% at 6 dBm. The higher measured efficiency at the large-power state is mainly due to the higher reverse breakdown voltage of the actual diode.

Table II displays a comparative analysis of the rectifier's performance compared with previous studies. The findings reveal that the proposed rectifier achieves a fractional bandwidth (with PCE > 50%) of 64.9%, the highest among the compared studies at 0 dBm, and it utilizes only one diode.

### IV. CONCLUSION

In this letter, a low-power and high-efficiency octave rectifier is proposed and designed for electromagnetic energy harvesting. The design is successfully broadened by partitioning the operating frequency band of the rectifier into two segments, resistance and impedance matching, in accordance with the impedance variation. The rectifier has a more than 50% rectification efficiency between 1.3 and 2.55 GHz. The rectifier design is applicable for electromagnetic energy harvesting.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \cos 2\theta_{CS} - \frac{1}{2}T_0Z_{CS}\sin 2\theta_{CS} & jZ_{CS}(\sin 2\theta_{CS} - T_0Z_{CS}\sin^2 \theta_{CS}) \\ j\frac{1}{Z_{CS}}(\sin 2\theta_{CS} + T_0Z_{CS}\cos^2 \theta_{CS}) & \cos 2\theta_{CS} - \frac{1}{2}T_0Z_{CS}\sin 2\theta_{CS} \end{bmatrix} \quad (5)$$

$$T_0 = \left( \frac{\tan \theta_{C1}}{Z_{C1}} + \frac{\tan \theta_{C2}}{Z_{C2}} \right) \quad (6)$$

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